SAMPLE

Earth's Dynamic Systems: Biomes

Earth's Major Ecosystems

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Introduction

Look at the planet from far enough away, as the whole earth shows itself to a satellite or a dreamer, and the land sorts itself: a green belt around the middle, dark forests in a wide northern ring, yellow grasslands where the rain falters, dun-colored desert cores, white tundra at the poles' edge. Those vast sweeps are not accidents of the map but living systems, each with its own climate, its own clothing of vegetation, its own animals and soils and habits of fire, and the science of ecology has a word for them, the biome: the world's major life zones, the largest communities of living things short of the whole planet, and the subject of this book. A biome is what you get when climate, over thousands of years, filters the world's plants down to a workable wardrobe, and the plants then build a landscape that everything else, from soil microbes to the largest herds, learns to live in: rainforest, steppe, prairie, desert, taiga, tundra, and the rest, each a solution, tested across whole continents, to the same handful of physical questions: how much water, how warm, how seasonal, how much light.

The book tells the story of the biomes at their own scale, continent-sized and planet-shaped, and the telling follows a plain arc. The early chapters build the idea: what a biome is and is not, where the concept came from, and how the master recipe works, the coupling of temperature and rainfall that decides, better than any other pair of numbers, which grand outfit a region wears. The middle chapters are the wardrobe itself, taken zone by zone: the tropical rainforest, the temperate and boreal forests, the world's grasslands, the deserts, the tundra, and the smaller but decisive systems at the edges, the Mediterranean scrublands and the wetlands. The later chapters widen to the questions that make the concept live: why boundaries between biomes blur and what that means; how fire, grazing, and time keep the zones moving; how freshwater and the ocean carry their own versions of the pattern; and how one species, converting grassland to field and forest to timber, has redrawn the map in a geological instant, at the very moment the climate that chose the wardrobe is itself being changed.

The telling keeps to plain language and asks the reader for no scientific background, only the patience to hold a large idea steadily: that the living world is organized, and that its organization is readable. The reward for that patience is a pair of eyes that see the planet differently afterward, able to read a change of scenery on any road, from the tropics outward or from the plains upward, as the visible logic of rainfall and warmth doing its patient work; and the book's quiet promise is that this logic, once learned and taken on the road, turns every landscape the reader ever crosses into a text worth reading slowly.

One paragraph of orientation, for the reader who prefers a route before the scenery. The first two chapters build the toolkit: the concept and its fences, then the master recipe of warmth and water that decides the wardrobe. Chapters three through nine are the wardrobe itself, the world's major life zones taken one at a time, from the equator's rainforest outward and poleward to the tundra, with the smaller decisive systems, the mediterranean scrublands and the wetlands, seated at the end of the sequence. The last four chapters widen the lens: the nature of the map's boundaries, the moving forces of fire and grazing and time, the water realm's parallel wardrobes, and the human hand that has redrawn everything. Nothing later needs anything earlier except attention, and the reader who wants one chapter at a sitting will find each stands on its own feet, though the wardrobe chapters do answer one another, as wardrobes will.

Chapter 1: What a Biome Is, and What It Is Not

The chapter begins with the definition, plainly stated, because precision at the start saves confusion later. A biome is a large region, typically spanning thousands to millions of square kilometers, defined not by the species living in it but by the dominant kind of life it carries: its characteristic vegetation, the growth forms and strategies of its plants, and the climate and soil that select them. Two biomes half a world apart may share almost no species, the grasslands of North America and the steppes of central Asia, the deserts of Arizona and the Karoo, and still count as the same biome, because the same physical conditions have driven evolution toward the same answers: drought-hardened leaves, fire-tolerant grasses, cold-adapted conifers, whatever the local inventories. The biome, in the phrase the textbooks lean on, is a community of characteristic life forms, not a club of particular species, and the distinction is the concept's whole power, letting the ecologist compare across continents as no species-based map could.

What the concept is not matters just as much, and the chapter's second section fences the idea in honestly. A biome is not an ecosystem, though every biome contains many: the ecosystem is the local machine of cycling energy and matter, stream, meadow, rotting log, while the biome is the grand class of machines that a region's climate makes possible. A biome is not a habitat, which is the address of a particular organism; a single biome

offers thousands of habitats, and some bold species, from geese to sharks to humans, commute across several biomes as if the map did not exist. And a biome is not a hard-edged province, despite the clean colors of the classroom map: the real zones grade into each other through wide transitional bands, a matter the book returns to further on, and the ecotone between two biomes is itself a distinctive strip of country with its own mix of conditions, often richer in species than either parent. The map's lines are honest simplifications of gradients, and the book keeps that honesty throughout.

The concept's history gives it a human shape, and the chapter's middle pages tell it briefly. The word arrived in the second decade of the twentieth century from an American plant ecologist who spent a career arguing that vegetation formed units, associations, that could be named and mapped like the units of geology; his word for the grandest units, biome, took hold slowly, surviving decades of debate over whether plant communities were real things or just human conveniences. The idea matured alongside the ecosystem concept, which a British botanist supplied in the 1930s by insisting that climate, soil, and organisms were parts of one interacting system, and it went global in the 1960s and 1970s, when a coordinated international program of ecological surveys, the largest the field had ever attempted, mapped the world's productivity and structure biome by biome, giving the concept its modern data and its modern authority. The classroom map of biomes is thus a young artifact, barely a century old in its idea and half a century old in its detail, and the science of the last two generations has been spent testing it, refining it, and occasionally, as later chapters record, fighting about it.

The chapter closes by fixing the scale hierarchy the reader will need for everything that follows, from the largest to the smallest: the biosphere, the sum of all life, which the biomes partition among themselves; the biome, defined by climate and dominant life forms, the book's subject; the ecosystem, the local machine; the community, its interacting populations; the population; and the individual organism, which experiences all the levels above as the weather and neighbors of its life. The biome sits high in that stack, and its study demands a special temperament, the ecologist's willingness to think in whole landscapes, and a special bookkeeping, which the next chapter begins: the accounting of warmth and water that decides, more than any committee of species ever could, what the world's great regions are permitted to wear.

One caution about scale belongs here, before the wardrobe chapters begin, because it prevents the commonest confusion. The biome is not the largest unit ecologists have drawn: above it sits the biogeographical realm, the grand continental isolation units whose histories the map of life on earth is really about, and whose borders, drawn where the oceans and deserts cut the land, mark where whole floras and faunas stopped mixing. What the realm contributes is cast, and what the biome contributes is costume: the same tropical rainforest recipe runs in the Neotropical realm, the Afrotropical, and the Indomalayan, and the three rainforests share a wardrobe but almost no actors, the jaguar, leopard, and jaguarundi substituting for one another in their costumes' sameness. That double hierarchy, realm for history and biome for climate, is what makes comparison across continents honest, and the wardrobe chapters lean on it throughout.